Factors, Multiples & Primes

Year 4 · **Homework Worksheet**

Worked Example

Worked example: Is 17 a prime number? Explain.

Answer: Yes — 17 can only be divided exactly by 1 and 17 itself, so it is a prime number.

Easy / Standard (15 questions)

1. Is 18 a multiple of 5?
2. Select ALL of these that are multiples of 2: A) 4 B) 7 C) 10 D) 15 E) 18
3. Is 5 a prime number or not? Explain briefly.
4. List all the factors of 21.
5. What is 7 squared?
6. Is 36 a multiple of 5? A) Yes, 36÷5=7.2 B) Yes, 36÷5=7 exactly C) No, 36÷5 is not exact D) Cannot tell
7. List the first five multiples of 3.
8. Is 3 a prime number or not? Explain briefly.
9. Which of these is NOT a factor of 24? A) 4 B) 6 C) 9 D) 8
10. What is 6 squared?
11. Is 26 a multiple of 10?
12. What is the 4th multiple of 6? A) 18 B) 20 C) 24 D) 30
13. Is 8 a prime number or not? Explain briefly.
14. Select ALL of these that are factors of 10: A) 1 B) 2 C) 3 D) 5 E) 10
15. What is 9 squared?

Standard (15 questions)

1. What is the lowest common multiple (smallest shared multiple) of 3 and 4?
2. Select ALL of these that are factor pairs of 40: A) 4 × 10 B) 5 × 8 C) 6 × 7 D) 2 × 20 E) 3 × 13
3. Find the common factors of 20 and 30.
4. Is 100 a square number? What is √100?
5. List all the prime numbers up to 31.
6. Find the LCM (smallest shared multiple) of 5 and 4. A) 9 B) 15 C) 20 D) 25
7. List all the factor pairs of 30.
8. Find the common factors of 16 and 24.
9. Is 64 a square number? What is √64? A) Yes; √64=6 B) Yes; √64=7 C) Yes; √64=8 D) No
10. List all the prime numbers up to 23.
11. What is the lowest common multiple (smallest shared multiple) of 5 and 2?

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12. Which of these is NOT a factor pair of 24? A) 1 × 24 B) 2 × 12 C) 4 × 6 D) 3 × 9

13. Find the common factors of 15 and 25.

14. Is 49 a square number? What is √49? A) Yes; √49=6 B) Yes; √49=7 C) Yes; √49=8 D) No

15. List all the prime numbers up to 57.

Standard / Advanced (15 questions)

1. Is 33 prime? If not, give its smallest factor other than 1.

2. Is 36 a square number? If so, what is its square root plus 4?

3. Find the HCF of 8 and 12, then multiply it by 3. A) 4 B) 8 C) 12 D) 16

4. Find the LCM of 3 and 4, then add 2.

5. List the factors of 20 that are also even numbers.

6. A square patio uses 36 tiles. The gardener wants to rearrange the same number of tiles into a rectangle that is not a square, using a whole number of rows. Give one possible rectangle arrangement, and explain why 36 is a square number.

7. A baker has 10 croissants and 15 muffins. She wants to put them into identical bags with the same number of croissants in each bag and the same number of muffins in each bag, using the greatest number of bags possible. How many bags can she make? A) 2 B) 3 C) 5 D) 10

8. Two flashing lights on a fairground ride blink every 2 seconds and every 3 seconds. They both blink together at the start. After how many seconds will they next blink together at exactly the same moment?

9. A code uses only prime numbers up to 12. List them, and give their sum.

10. A teacher has 24 pupils and wants to arrange them into equal rows with more than 1 pupil per row and more than 1 row. Which of these are valid row arrangements? Select ALL that apply. A) 2 rows of 12 B) 3 rows of 8 C) 5 rows of 5 D) 4 rows of 6 E) 7 rows of 3

11. A baker has 18 croissants and 12 muffins. She wants to put them into identical bags with the same number of croissants in each bag and the same number of muffins in each bag, using the greatest number of bags possible. How many bags can she make, and how many croissants and muffins go in each bag?

12. Two flashing lights blink every 4 seconds and every 6 seconds, both blinking together at the start. After how many seconds will they next blink together at exactly the same moment? A) 4 B) 6 C) 10 D) 12

13. A square patio uses 25 tiles. The gardener wants to rearrange the same number of tiles into a rectangle that is not a square, using a whole number of rows. Give one possible rectangle arrangement, and explain why 25 is a square number.

14. A code uses only prime numbers up to 15. List them, and give their sum.

15. A teacher has 20 pupils and wants to arrange them into equal rows with more than 1 pupil per row and more than 1 row. Which of these are valid row arrangements? Select ALL that apply. A) 2 rows of 10 B) 4 rows of 5 C) 3 rows of 7 D) 5 rows of 4 E) 6 rows of 3

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